Sharzil Hasnine

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Education

Bachelor of Computer Science (Advanced) Feb 2023 – Nov 2026
Adelaide University

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, C++, C, Bash, MATLAB, GDScript
Libraries and Frameworks: Node.js, React, Spring Boot, REST APIs, MySQL, MongoDB, Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, ROS2, Nav2, Gazebo, Godot
Tools & Practice: Git, Docker, Linux, Jira, JUnit, automated testing, CI/CD

Experience

Student Data Engineer - REDCap-KNIME Integration - Adelaide University Aug 2026 - Oct 2026
Tech: *Python, KNIME, REDCap*

- Connected a research data system (REDCap) to an analytics workflow tool (KNIME) by building a custom Python extension using PyCap to pull records and metadata through the REST API into separate typed output tables.
- Implemented metadata-driven ETL and validation: decoded categorical and checkbox fields, enforced field-level validation rules, and preserved longitudinal event and repeating-instrument context.
- Built a controlled test environment of 10,000 synthetic records across 64 fields (no real participant data), enabling reproducible validation of the extraction and ETL workflow.

Industry Project - Block Model Compression Algorithm - Maptek Aug 2025 - Oct 2025
Tech: *C++, Bash, Jira*

- Architected a stream-oriented lossless compression system for 3D data, optimising for speed and compression ratio.
- Implemented an octree-based spatial compression algorithm, achieving 98.8% reduction in data size while improving processing speed by 2.6x over the uncompressed baseline, forming the foundation for subsequent team optimisations reaching 99.17% compression and 1520x processing speed.
- Designed a testing framework that verifies correctness of compressed outputs against uncompressed input data, ensuring algorithmic correctness independent of compression strategy and enabling automated, extensible testing.
- Built a cross-platform CI pipeline with automated testing and reproducible Linux and Windows builds, integrating cross-compilation and artifact generation into the development workflow, improving team velocity by 1.5x.

Software Developer - Autonomous Navigation - Adelaide Rover Team Oct 2024 - Feb 2025
Tech: *Python, C++, ROS2, Nav2, Gazebo, Groot*

- Designed a behavior tree for the rover's autonomous navigation pipeline using Nav2, enabling modular decision-making for tasks such as path planning, recovery behaviours, and goal execution.
- Simulated navigation behaviour using Gazebo, supporting the Adelaide Rover Team's preparation for the Australian Rover Challenge's autonomous navigation task.

Projects

Daily Iftar Registration System Feb 2026 - Mar 2026
Tech: *TypeScript, Node.js, MongoDB, REST API, Docker*
Link: *aumsa.com.au*

- Delivered production-ready backend features under tight release cycles, building and iterating on RESTful APIs (Node.js, MongoDB) to support live event registration used by 70+ daily users.
- Architected DST-safe, timezone-aware date handling (Australia/Adelaide), eliminating off-by-one scheduling bugs and ensuring reliable, real-world consistency for day-based operations.
- Designed scalable MongoDB database schemas and API contracts to support rapid feature expansion and seamless frontend integration, maintaining stability while continuously deploying new functionality.

Organisation and Event Management Website Apr 2024 - Jun 2024
Tech: *MySQL, REST API, Node.js, Javascript, React*

- Designed and implemented the MySQL relational schema for a platform where organisations publish events, users sign up for them, and users follow organisations to see their updates on a personalised homepage.

- Built the REST API endpoints that serve this data, including event sign-ups and the follow-based homepage feed.

Paxos Consensus Algorithm Sep 2025 - Oct 2025

Tech: *Java, JUnit, Bash*

Link: *github.com/Shas-9/Paxos*

- Implemented the Paxos Consensus Algorithm without use of any external libraries, enabling multiple concurrent nodes to elect a single leader despite network delays and node failures.
- Developed a TCP socket-based messaging protocol supporting Paxos phases (prepare, promise, accept, learn) with nodes dynamically acting as proposers, acceptors, and learners to achieve consensus.
- Designed a failure simulation framework introducing latency, message loss, and node crashes to validate consensus correctness under concurrent proposals and partial system failures.

Distributed Weather Feed Aggregator with Lamport Clocks Aug 2025 - Sep 2025

Tech: *Java, JUnit, REST API*

Link: *github.com/Shas-9/Weather-Aggregation-Server*

- Engineered a distributed data aggregation system supporting concurrent clients and content servers.
- Implemented Lamport logical clocks to maintain causal consistency and deterministic ordering of concurrent updates across distributed components, without use of external libraries.
- Developed a multi-threaded aggregation server with thread-safe request processing, persistent storage with crash recovery and automated 30s expiry of stale data sources, ensuring system reliability under concurrent workloads.

SVM and Neural Network Implementation Aug 2025 - Oct 2025

Tech: *Python, NumPy, Sklearn*

Link: *github.com/Shas-9/SVM-and-Neural-Network-Implementation*

- Implemented a Support Vector Machine from scratch in Python by solving the dual optimization problem with quadratic programming, including RBF kernel support and hyperparameter tuning using validation data.
- Built a two-layer neural network from scratch using forward and backward propagation, implementing core modules (Linear, ReLU, Cross-Entropy) and training with mini-batch SGD on a dataset with 200 features and 8.5k samples.
- Developed an end-to-end ML training pipeline including data standardization, train/validation/test splits, hyperparameter search, and performance evaluation using accuracy metrics with Sklearn.

CatQuest – 2D Maze Exploration Game Aug 2023 - Oct 2023

Tech: *C++, SFML*

Link: *github.com/Shas-9/Labyrinth-game*

- Engineered modular game architecture, separating game state, entities, and UI for maintainability and scalability.
- Integrated enemy AI and player mechanics within the existing procedural random maze generation system, including real-time collision handling and health management.
- Designed the game UI, with seamless navigation and state transitions without affecting game loop performance.

Extracurriculars

Adelaide University Competitive Programming League (AUCPL) Mar 2026 – Oct 2026

- Achieved a team 2nd place finish in the Adelaide University Competitive Programming League.

Secretary Nov 2024 – Aug 2026

ISSUA

- Led internal communications for a 10+ member committee, producing structured meeting minutes and action item documentation to ensure organizational alignment and accountability.
- Managed university-facing logistics including room bookings, building access, and prayer room maintenance.
- Administered organizational knowledge management using Google Drive, maintaining structured repositories of meeting records, event documentation, and operational resources to support cross-functional collaboration.

Events Officer Nov 2023 – Nov 2024

ISSUA

- Planned and executed society events for 100+ attendees, overseeing end-to-end logistics, budget management, and stakeholder alignment to meet organizational objectives.
- Coordinated event setup, including venue preparation, infrastructure, and audio-visual systems.
- Engaged with attendees and guests during events, providing support and fostering a welcoming environment.